Sebastian Tamayo

Senior Software Engineer

+57 3113698653• sebstiantamayo629@outlook.com• https://www.linkedin.com/in/sebastian-tamayo-pro/ • Pereira, Colombia

Seasoned Senior Web Developer with over a decade of experience architecting, developing, and optimizing high-performance, scalable web applications. Expert in crafting dynamic, mobile-first UIs using Angular, and building robust, secure backend services with Node.js, Express.js, and RESTful APIs.

Proficient across the entire software development lifecycle, including modular system design, relational and NoSQL data modeling, CI/CD automation, and cloud-native deployments with AWS, Docker, and Kubernetes. Proven ability to thrive in Agile/Scrum environments, collaborating closely with cross-functional teams to deliver impactful, business-aligned solutions.

Recognized for writing clean, maintainable, and testable code, with a strong focus on SOLID principles, Clean Architecture, and Test-Driven Development (TDD). Skilled in Angular performance optimization, component-based architecture, and Node.js backend engineering—including API design, caching strategies with Redis, and asynchronous event-driven systems using RabbitMQ or Kafka.

A lifelong learner driven by innovation, with deep expertise in the JavaScript/TypeScript ecosystem, modern DevOps practices, and a passion for delivering production-ready software that enhances user experience and drives measurable results.

PROFESSIONAL EXPERIENCE

UST

Senior Software Engineer ***06.2022 – 06.2025***

- Collaborated within a cross-functional Agile Scrum team, mentoring junior developers and conducting rigorous code reviews with an emphasis on SOLID principles, clean architecture, and modular, testable design across both frontend and backend codebases.
- Rebuilt a high-traffic, customer-facing web application using Angular, delivering a responsive, mobile-first UI with dynamic data binding, real-time updates via WebSockets, and WCAG-compliant accessibility. Integrated AWS IVS for live streaming and connected with Shopify Storefront API, resulting in higher user engagement and improved conversion rates.
- Developed scalable, secure backend services using Node.js and Express.js, building modular RESTful APIs protected with JWT-based authentication. Modeled complex domain logic with Sequelize and Mongoose, using PostgreSQL and DynamoDB for both relational and NoSQL use cases.
- Designed a microservices-inspired backend architecture, containerized with Docker and deployed to Amazon ECS, supporting asynchronous workflows and event-driven communication via Apache Kafka and RabbitMQ, with real-time features powered by Socket.IO and Server-Sent Events (SSE).
- Spearheaded the development of a production-ready AI-powered chatbot using AWS Lex, AWS Lambda, and Amazon DynamoDB, supporting intent-based conversation flows and reducing customer service costs through automation and smart fallback logic.
- Optimized Angular frontend performance using lazy loading, preloading strategies, and build optimizations via Webpack and Vite. Backend performance was enhanced through query optimization, Redis caching, and asynchronous task execution with BullMQ and Node worker threads, resulting in a 30% improvement in speed and scalability.
- Architected and maintained CI/CD pipelines using GitHub Actions and Jenkins, integrating automated testing tools including Jest, Cypress, Playwright, and Postman. Reduced regression risks and improved deployment velocity through test coverage enforcement and build step validations.

Tech Stack: Angular, RxJS, NgRx (Akita), TypeScript, HTML5, SCSS, Node.js, Express.js, REST API, JWT, PostgreSQL, Redis, DynamoDB, Apache Kafka, RabbitMQ, WebSockets (Socket.IO), SSE, Shopify Storefront API, Docker, AWS, GitHub Actions, Jenkins, Jest, Cypress, Playwright, Postman, Webpack, Vite, SEO, WCAG, Figma, TDD

Gorilla Logic

Full Stack Engineer *02. 2018 – 05.2022*

- Designed and implemented responsive, mobile-first user interfaces using Angular, TypeScript, and SCSS, leading to a 40% increase in mobile user retention and a 30% boost in platform-wide engagement.
- Developed scalable and reusable UI component libraries for enterprise-grade dashboards, leveraging RxJS and NgRx (or Akita) for reactive state management—accelerating development velocity and promoting UI consistency across teams.
- Built secure, modular REST APIs using Node.js and Express.js, aligned with domain-driven design and clean architecture principles. Implemented JWT-based authentication, custom middleware, and structured response serializers to enforce API integrity.
- Integrated real-time features such as live alerts and streaming dashboards using WebSockets (Socket.IO) and Server-Sent Events (SSE), dramatically improving interactivity and responsiveness for time-sensitive applications.
- Containerized backend services using Docker, and deployed them on AWS ECS, with infrastructure provisioned and scaled using AWS CloudFormation, resulting in faster, more reliable deployments and simplified DevOps workflows.
- Tuned PostgreSQL databases for high performance using query profiling, advanced indexing, and data normalization techniques—achieving a 15% reduction in API latency under high traffic loads.
- Architected asynchronous, event-driven systems with Apache Kafka and RabbitMQ, enhancing fault tolerance and horizontal scalability in distributed data processing pipelines.
- Applied Test-Driven Development (TDD) with Jest, Cypress, and Factory Boy equivalents, while establishing robust observability stacks using Prometheus, Grafana, Elastic APM, and AWS CloudWatch, ensuring proactive monitoring and fast incident resolution.

Tech Stack: Angular, TypeScript, RxJS, NgRx (Akita), HTML5, SCSS, Node.js, Express.js, REST API, JWT, PostgreSQL, Redis, Apache Kafka, RabbitMQ, WebSockets (Socket.IO), SSE, Docker, AWS (ECS, Lambda, S3, CloudFormation, CloudWatch), Prometheus, Grafana, Elastic APM, Git, GitHub Actions, Jenkins, Jest, Cypress, Postman, Jasmine, Protractor, Figma, SEO, WCAG, TDD

Sophos Solutions

Senior Web Developer *09.2015 – 02.2018*

- Spearheaded the development of a high-performance Single Page Application (SPA) for a financial services platform using Angular, RxJS, and NgRx (or Akita), achieving a 30% increase in user engagement and significantly improving client-side rendering speed.
- Modernized a legacy monolith by decoupling the frontend from the backend, integrating Angular with secure, scalable Node.js + Express.js REST APIs, and implementing JWT-based authentication, leading to a 40% reduction in server response time and increased backend modularity.
- Enhanced frontend performance with techniques such as code splitting, lazy loading, tree shaking, and asset optimization. Improved SEO through structured metadata, Open Graph, and JSON-LD, resulting in a 35% improvement in page speed scores.
- Delivered fully responsive, mobile-first UIs adhering to WCAG 2.1 accessibility standards and optimized for Google AMP, contributing to a 25% increase in mobile traffic and improved Lighthouse performance metrics.
- Implemented an end-to-end testing strategy using Jasmine, Protractor, Cypress, and Jest, fully integrated into CI/CD pipelines via GitHub Actions and Jenkins, which reduced post-release defects by over 75%.
- Collaborated closely with UI/UX designers using Figma and Adobe XD to deliver modular, reusable Angular components for real-time financial dashboards, boosting render performance by 50% and enabling faster parallel development across teams.

Tech Stack: Angular, TypeScript, RxJS, NgRx (Akita), HTML5, SCSS, Node.js, Express.js, REST API, JWT, MySQL, Redis, Webpack, Vite, Docker, AWS (ECS, Lambda, S3, CloudFormation, CloudWatch), GitHub Actions, Jenkins, Jasmine, Protractor, Cypress, Jest, Playwright, Postman, Google AMP, WCAG, SEO, Open Graph, JSON-LD, Google Analytics, Adobe XD, Figma, CI/CD, TDD

Web Developer ***09.2012 – 02.2015***

- Developed highly responsive, WCAG 2.1-compliant web applications using Angular, TypeScript, HTML5, and SCSS, ensuring seamless cross-browser compatibility and improving user engagement across both mobile and desktop platforms.
- Engineered scalable and maintainable frontend architectures by designing modular Angular components integrated with RxJS, Angular Router, and NgRx (or Akita), adhering to enterprise UI/UX standards and facilitating smooth product evolution.
- Designed and implemented secure, RESTful backend services using Node.js, Express.js, and JWT authentication, supporting complex business logic, workflow orchestration, and role-based access control for enterprise applications.
- Optimized build and deployment workflows with Webpack and Vite, and automated environment-specific configuration to streamline CI/CD delivery and ensure efficient asset bundling for production environments.
- Enhanced system performance and scalability by integrating Redis for caching and BullMQ or RabbitMQ for background task processing, reducing latency and ensuring stability under high traffic.
- Tuned PostgreSQL and MySQL databases by applying advanced query profiling, index optimization, and ORM-level performance tuning using TypeORM or Sequelize, resulting in improved data throughput and lower query response times.
- Increased frontend performance by over 30% through techniques such as code splitting, lazy loading, image optimization, and asset minification, positively impacting SEO rankings and Lighthouse performance metrics.
- Established a comprehensive testing strategy using Jasmine, Protractor, Jest, and Cypress, fully integrated into GitHub Actions and Jenkins CI/CD pipelines to support continuous integration, quality assurance, and deployment efficiency.

Tech Stack: Angular, TypeScript, RxJS, NgRx (Akita), HTML5, SCSS, Node.js, Express.js, REST API, JWT, PostgreSQL, MySQL, Redis, BullMQ, RabbitMQ, TypeORM, Sequelize, WebSockets, Server-Sent Events (SSE), Webpack, Vite, GitHub Actions, Jenkins, Azure App Service, Jasmine, Protractor, Jest, Cypress, Postman, Figma, Google Analytics, SEO, WCAG, CI/CD, TDD

EDUCATION

Universidad Católica de Pereira 04.2005 – 09.2012

Bachelor of Computer Science

SKILLS

Languages:	C++, JS(ES6), TS, Python, C#, Rust, PHP, Java, HTML5
Frameworks and Libraries:	React.js, Next.js, Gatesby, Angular, Node.js, Express, React Native, Redux, RxJS, REST API, Django, JavaSpringBoot, Antd, GraphQL, Ably, Laravel, Tailwind CSS, RabbitMQ, Kafka, TypeORM, Shopify, WordPress
DBMS	MySQL, MS SQL, PostgreSQL, MongoDB, AWS DynamoDB, AWS RDS, Azure SQL Database
IDEs and Tools:	Visual Studio Code, Jenkins, Cypress, Jest, Jasmine, Terraform, Docker, Kubernetes, Redux DevTools, Babel, Webpack, Storybook, Azure DevOps, Git Bash, PowerShell
Cloud Services:	AWS Cloud Services, Microsoft Azure, GCP
Other skills:	Problem Solving, Full Stack Development, Critical Thinking, Leadership, Algorithm Design, Team-Oriented, SOLID, Agile (Scrum), Microservices, Test-Driven Development (TDD), Diagnosing, Business Analysts